

Task 1

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Librarys:

```
library(tidyverse)
library(ggplot2)
```

Set WD and import data

```
gdp_per_capita_df <- read.csv("C:/Users/trabu/OneDrive/Documenti/Desktop/ids_k9/gdp_per_capita_df.csv")
head(gdp_per_capita_df) %>%
  filter(Year >= 2015)
```

```
##      Entity Year Code Gdp_per_capita Continent      Growth Ldc Population
## 1 Afghanistan 2015  AFG      2108.714      Asia -1.6664275  TRUE   33831764
## 2 Afghanistan 2016  AFG      2101.422      Asia -0.3458032  TRUE   34700612
## 3 Afghanistan 2017  AFG      2096.093      Asia -0.2535902  TRUE   35688935
## 4 Afghanistan 2018  AFG      2060.699      Asia -1.6885701  TRUE   36743039
## 5 Afghanistan 2019  AFG      2079.922      Asia  0.9328340  TRUE   37856121
## 6 Afghanistan 2020  AFG      1968.341      Asia -5.3646678  TRUE   39068979
##      Gdp
## 1 71341514391
## 2 72920629470
## 3 74807326831
## 4 75716343724
## 5 78737775117
## 6 76901073194
```

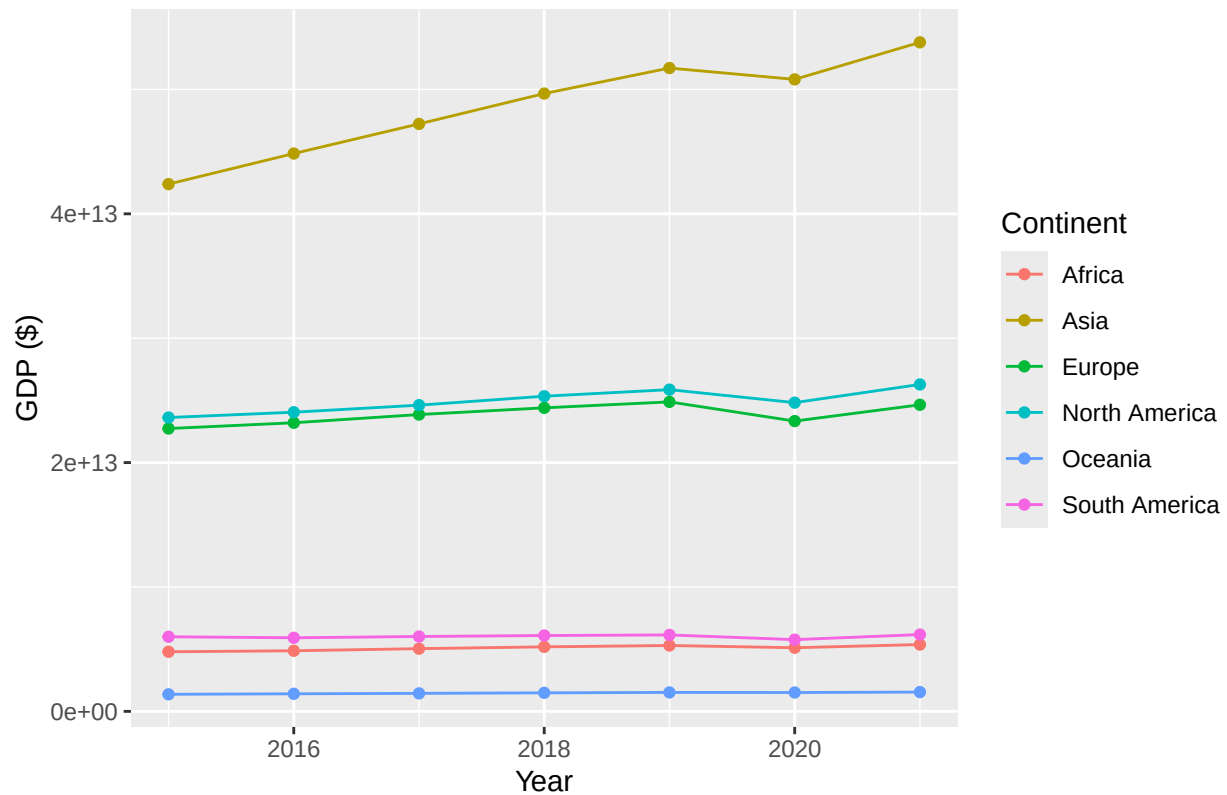
Calculate the total GDP of each continent

```
continent_gdp_totals <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Total_GDP = sum(Gdp, na.rm = TRUE)
  ) %>%
  ungroup()
```

Plot a graph of GDP for each continent, vs time

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = Total_GDP, colour = Continent)) +
  geom_point() +
  geom_line() +
  labs(title = "GDP/Year for 6 continents", y = "GDP ($)")
```

GDP/Year for 6 continents



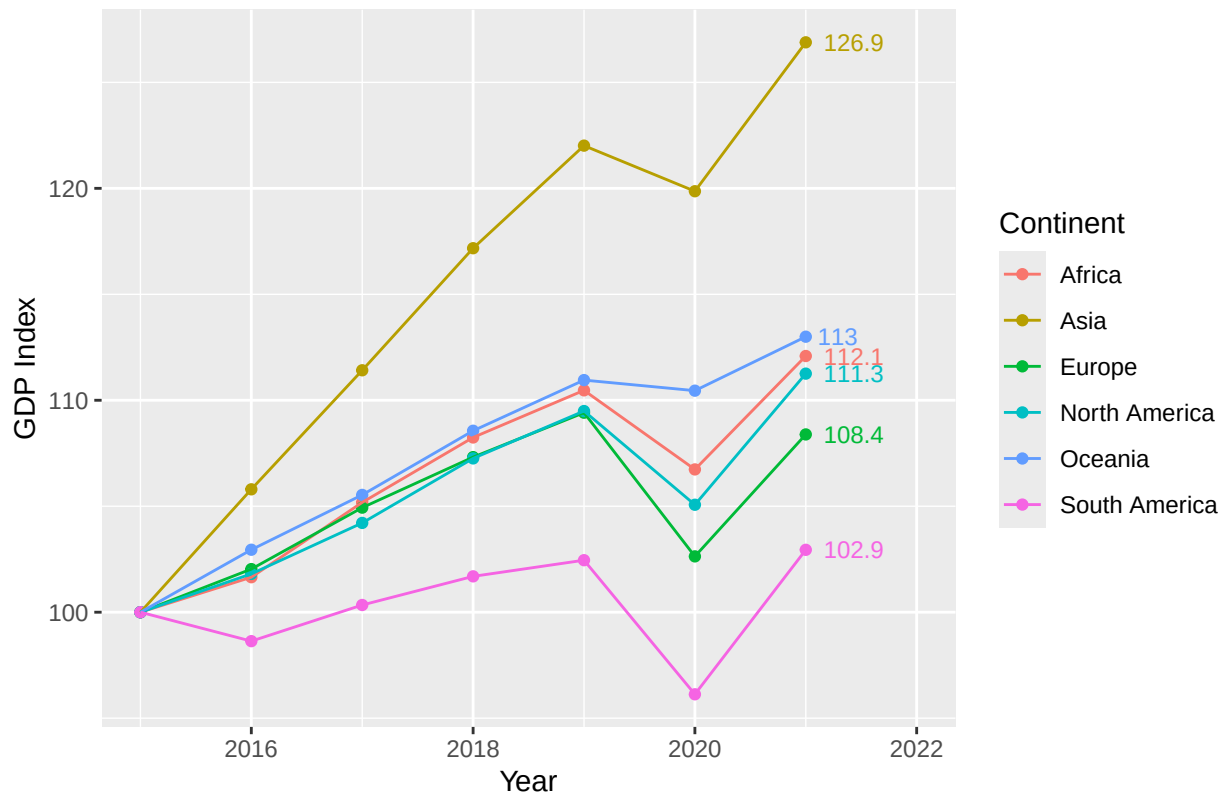
I decided this graph could show more if they all started at an index of 100

```
continent_gdp_totals <- continent_gdp_totals %>%
  group_by(Continent) %>%
  mutate(
    GDP_Index = Total_GDP / first(Total_GDP) * 100
  )
```

We now replot the graph, using normalised values

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = GDP_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_text(
    data = continent_gdp_totals %>% filter(Year == 2021),
    aes(label = round(GDP_Index, 1)),
    hjust = -0.3,
    size = 3,
    show.legend = FALSE) +
  expand_limits(x = 2022) +
  labs(title = "GDP Index/Year for 6 continents", y = "GDP Index")
```

GDP Index/Year for 6 continents



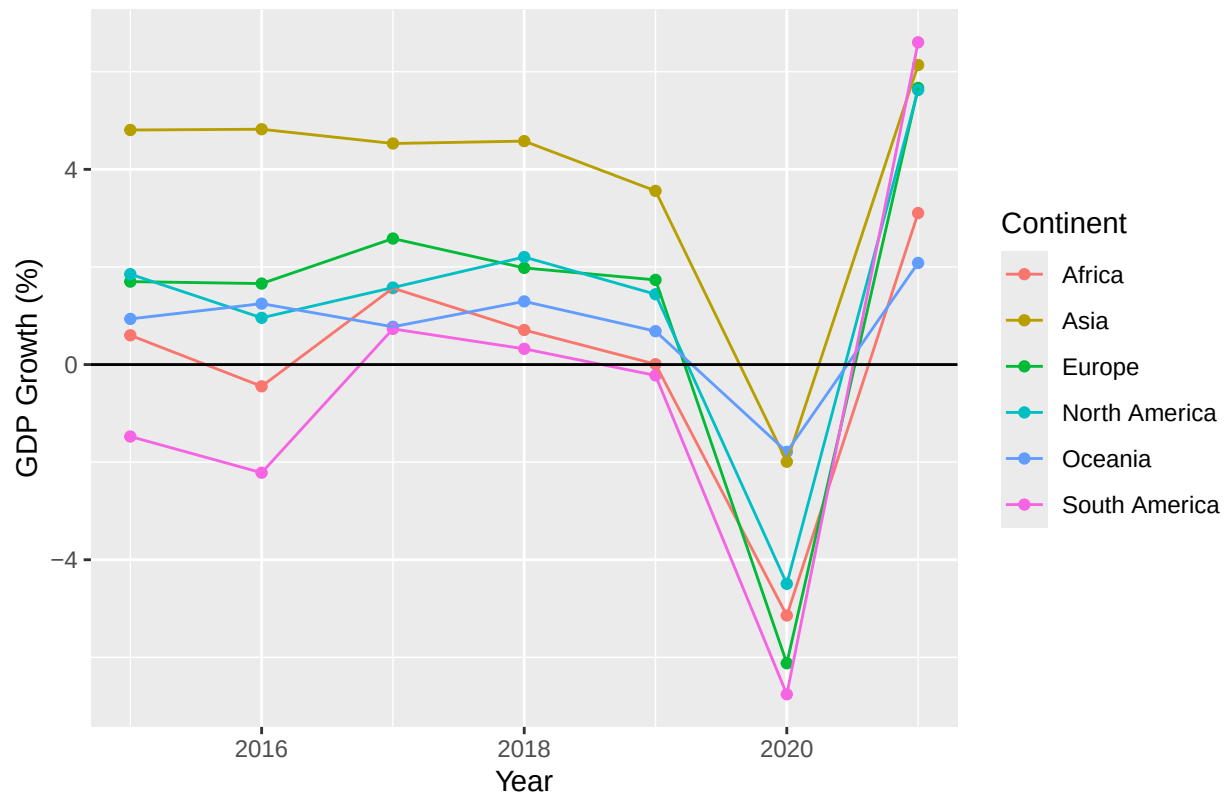
Calculate the weighted and unweighted growth of each continent

```
continents_growth_df <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Weighted_Growth = sum(Growth * Gdp, na.rm = TRUE) / sum(Gdp, na.rm = TRUE),
    Avg_Growth = mean(Growth, na.rm = TRUE)
  ) %>%
  ungroup()
```

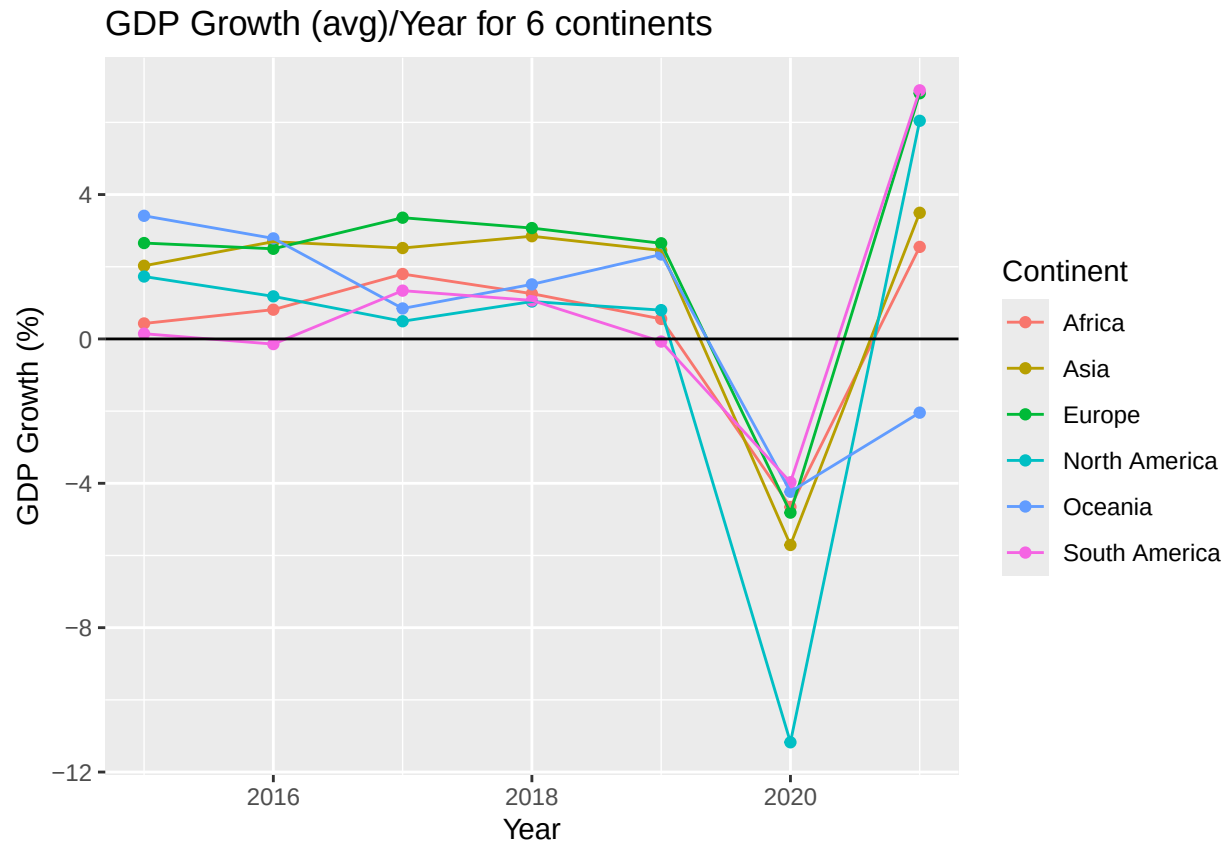
Plot the growth graphs of each continent

```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```

GDP Growth (weighted)/Year for 6 continents



```
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth (avg)/Year for 6 continents", y = "GDP Growth (%)")
```

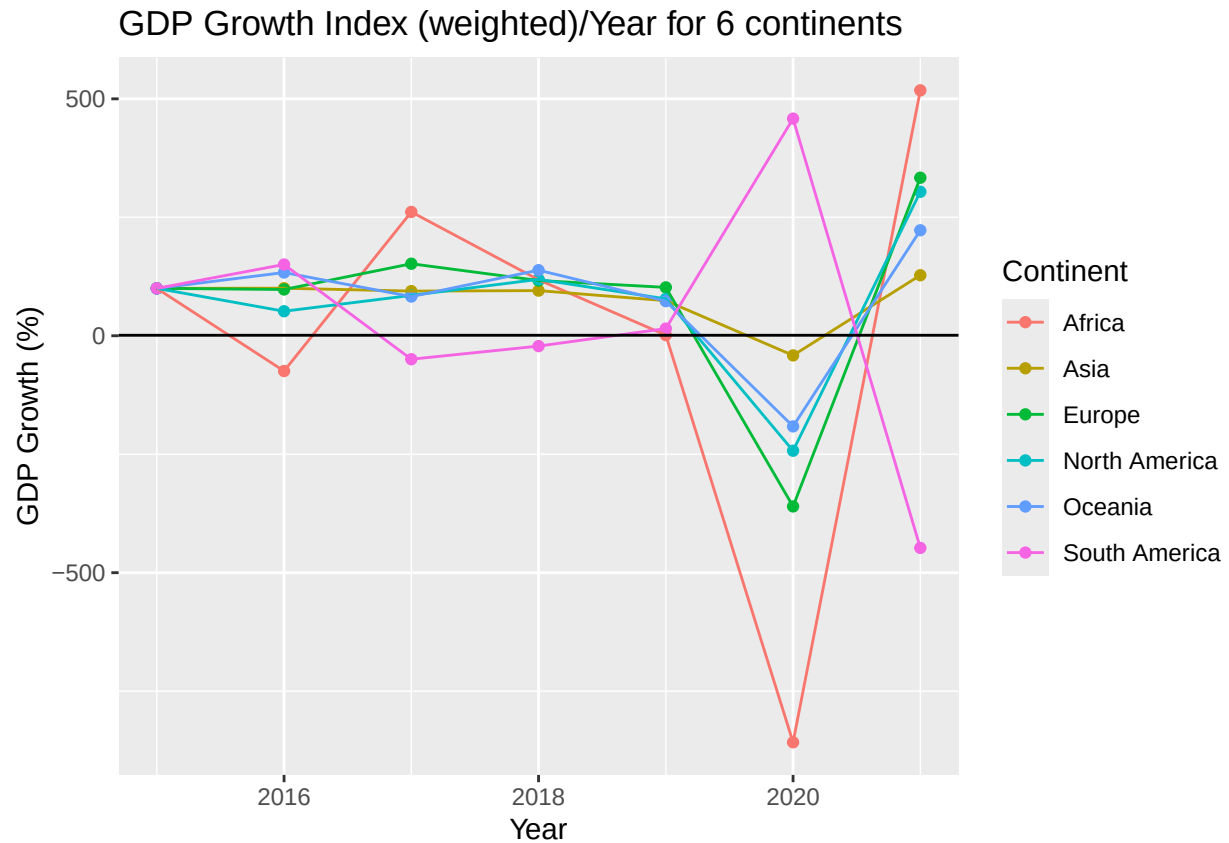


I decided to do a similar thing with growth about index (realised is redundant, don't use in analysis)

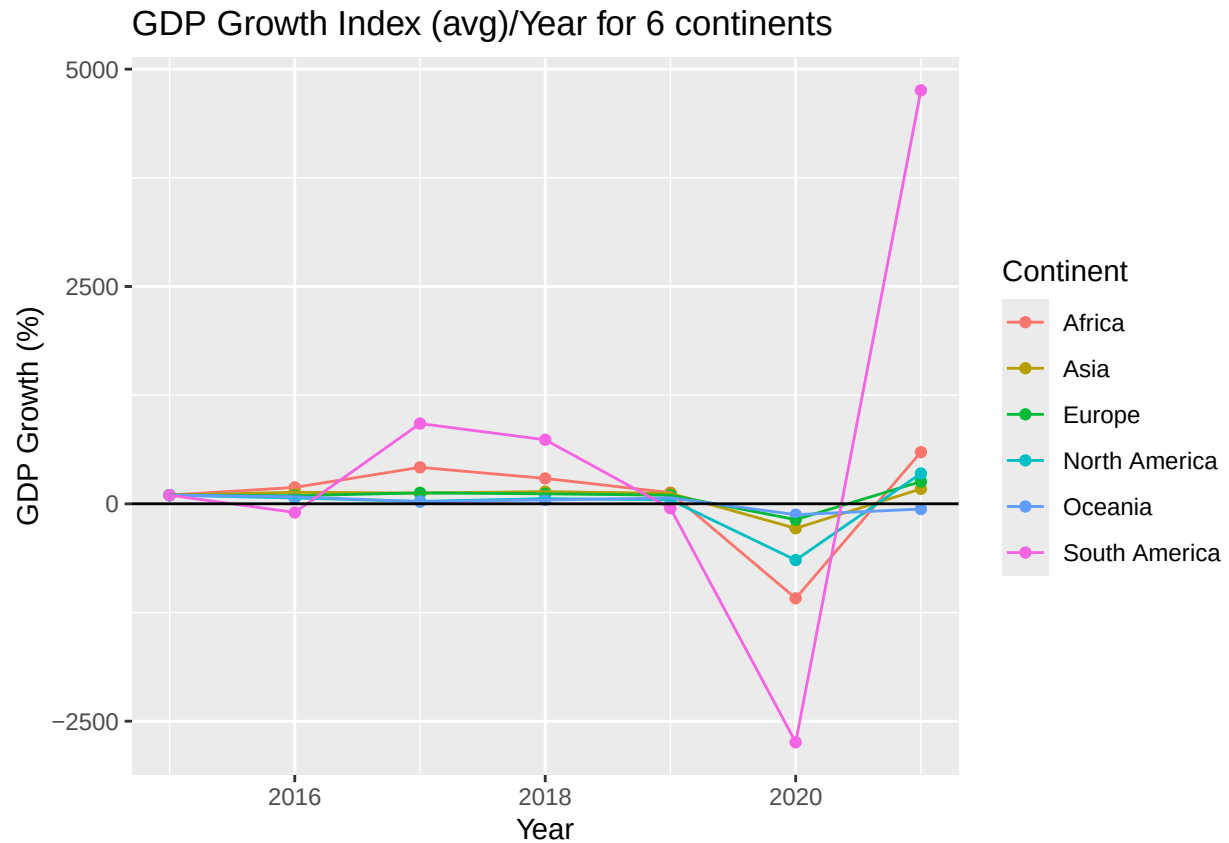
```
continents_growth_df <- continents_growth_df %>%
  group_by(Continent) %>%
  mutate(
    Avg_Growth_Index = Avg_Growth / first(Avg_Growth) * 100,
    Weighted_Growth_Index = Weighted_Growth / first(Weighted_Growth) * 100
  )
```

Plotting the graphs

```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 1) +
  labs(title = "GDP Growth Index (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```



```
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 1) +
  labs(title = "GDP Growth Index (avg)/Year for 6 continents", y = "GDP Growth (%)")
```



REF:

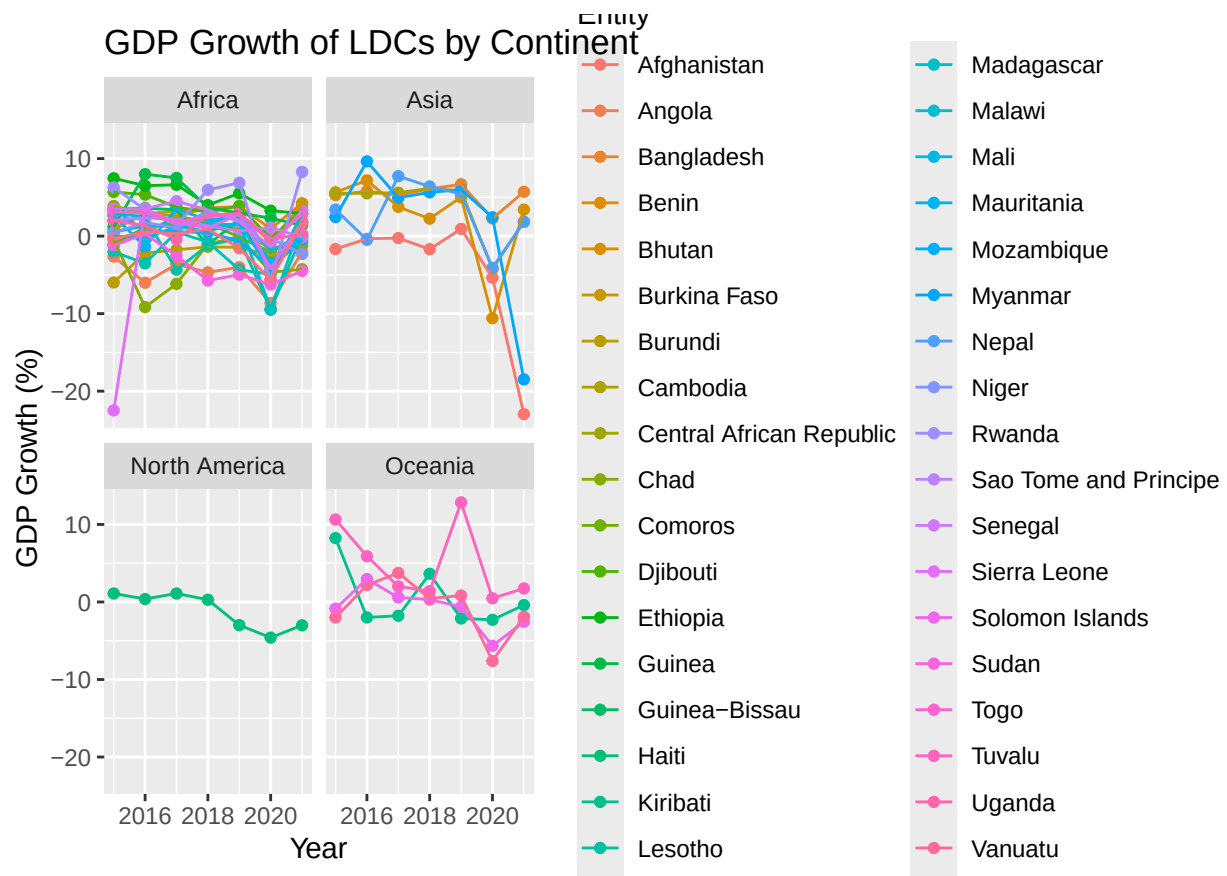
28/11/2025 - asked ChatGPT - "i have text near the last points on the x axis, how can i make it so it doesn't overlap the axis"

Create an LDC database

```
ldc <- gdp_per_capita_df %>%
  filter (Ldc == TRUE)
```

Plot GDP growth by continent

```
ggplot(ldc, aes(x = Year, y = Growth, color = Entity)) +
  geom_point() +
  geom_line() +
  facet_wrap(~ Continent) +
  labs(
    title = "GDP Growth of LDCs by Continent",
    x = "Year",
    y = "GDP Growth (%)"
  )
```



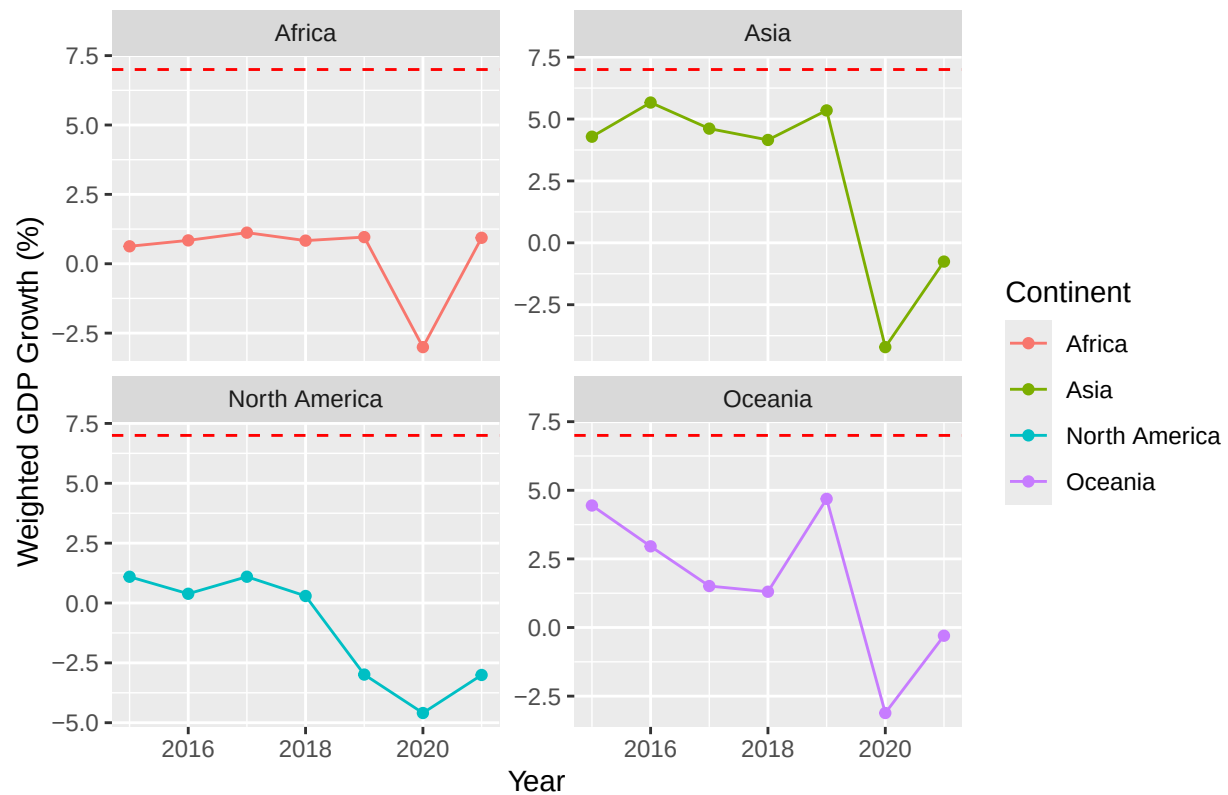
Compute the weighted average GDP growth for each continent

```
ldc_weighted <- ldc %>%
  group_by(Year, Continent) %>%
  summarise(
    weighted_growth =
      sum(Gdp_per_capita * Growth, na.rm = TRUE) /
      sum(Gdp_per_capita, na.rm = TRUE),
    .groups = "drop"
  )
```

Plot weighted GDP growth by continent with a 7% reference line

```
ggplot(ldc_weighted, aes(x = Year, y = weighted_growth, color = Continent)) +
  geom_point() +
  geom_line() +
  facet_wrap(~ Continent, scales = "free_y") +
  geom_hline(yintercept = 7, linetype = "dashed", color = "red") +
  labs(
    title = "Weighted GDP Growth of LDCs by Continent",
    x = "Year",
    y = "Weighted GDP Growth (%)"
  )
```

Weighted GDP Growth of LDCs by Continent



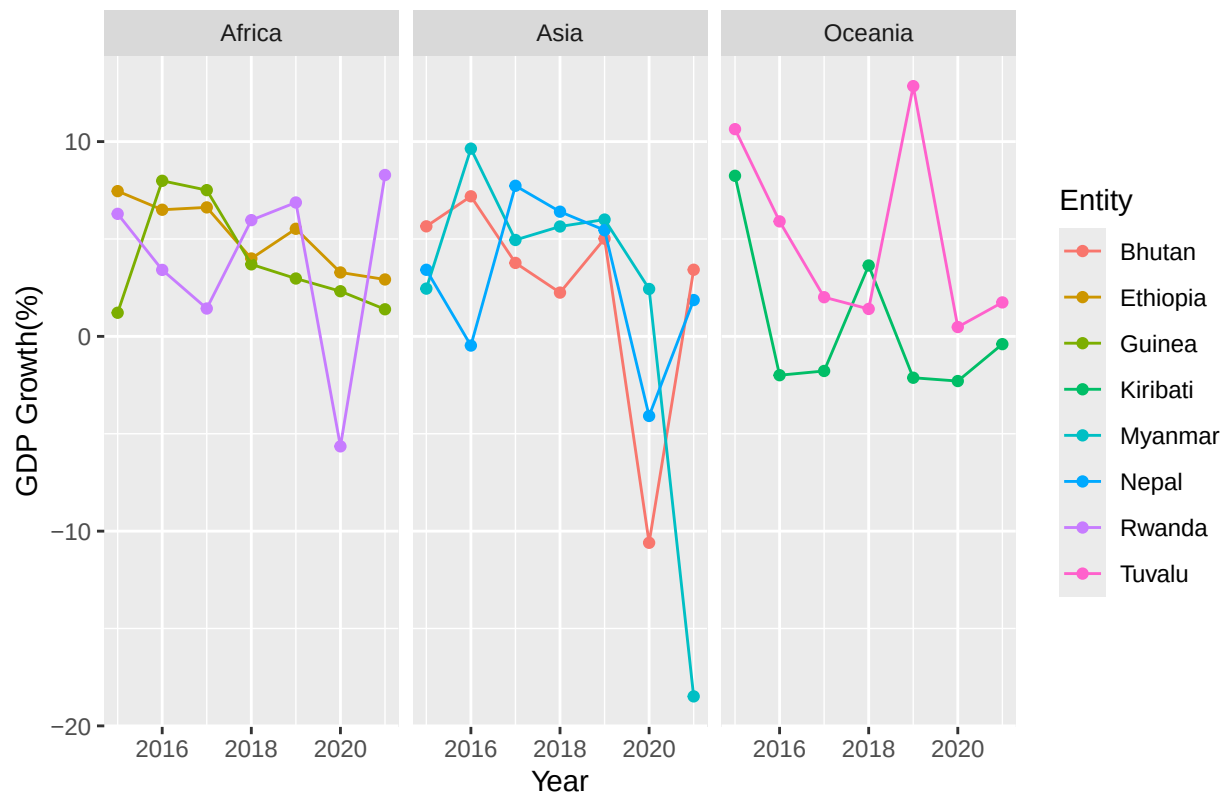
Identify which LDCs have had a growth that reached 7% and plot their growth

```
ldc_7 <- ldc %>%
  group_by(Entity) %>%
  filter(any(Growth >= 7)) %>%
  ungroup()
```

Plot their graph

```
ggplot(ldc_7, aes(x = Year, y = Growth, color = Entity)) +
  geom_point() +
  geom_line() +
  facet_wrap(~ Continent) +
  labs(
    title = "GDP Growth of LDCs that reached 7%",
    x = "Year",
    y = "GDP Growth(%)"
  )
```

GDP Growth of LDCs that reached 7%



Determine which countries have had the highest growth over the years

```
ldc %>%
  group_by(Entity) %>%
  summarise(avg_growth = mean(Growth, na.rm = TRUE)) %>%
  arrange(desc(avg_growth))
```

```
## # A tibble: 38 x 2
##   Entity      avg_growth
##   <chr>      <dbl>
## 1 Bangladesh    5.30
## 2 Ethiopia      5.19
## 3 Tuvalu        5.00
## 4 Guinea        3.87
## 5 Rwanda        3.80
## 6 Cambodia      3.75
## 7 Djibouti      3.54
## 8 Nepal         2.90
## 9 Senegal       2.67
## 10 Burkina Faso  2.41
## # i 28 more rows
```

REF:

28/11/2025 - asked ChatGPT - “I have a facet plot in ggplot with some continents having tons of lines and some only one. How can I adjust the y-axis scales so each graph uses the right space for its lines and they don’t look squished or empty?”

Find the 3 countries with most GDP growth, and 3 countries with most GDP decline

```
# Top 3 growth
top_growth <- gdp_per_capita_df %>%
  group_by(Entity) %>%
  summarise(Avg_Growth = mean(Growth, na.rm = TRUE)) %>%
  arrange(desc(Avg_Growth)) %>%
  slice(1:3)

# Top 3 decline
top_decline <- gdp_per_capita_df %>%
  group_by(Entity) %>%
  summarise(Avg_Growth = mean(Growth, na.rm = TRUE)) %>%
  arrange(Avg_Growth) %>%
  slice(1:3)
```

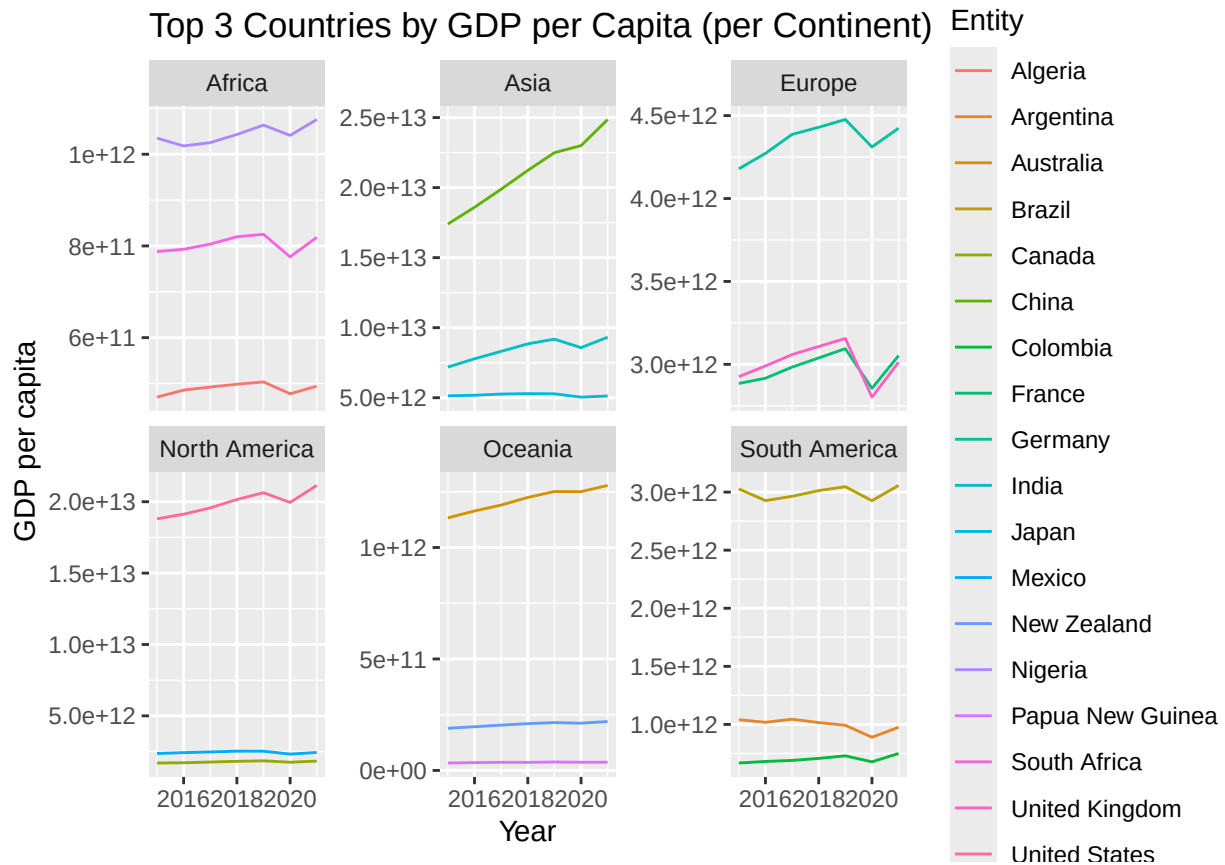
Find the countries with biggest gdp in each continent

```
top_3_per_continent <- gdp_per_capita_df %>%
  group_by(Continent, Entity) %>%
  summarise(Total_GDP = sum(Gdp, na.rm = TRUE)) %>%
  arrange(Continent, desc(Total_GDP)) %>%
  group_by(Continent) %>%
  slice(1:3)
```

Plot their graphs

```
filtered_df <- gdp_per_capita_df %>%
  semi_join(top_3_per_continent, by = c("Continent", "Entity"))

ggplot(filtered_df, aes(x = Year, y = Gdp, colour = Entity)) +
  geom_line() +
  facet_wrap(~ Continent, scales = "free_y") +
  labs(title = "Top 3 Countries by GDP per Capita (per Continent)", y = "GDP per capita")
```



#free y scaling - allows the y-axis of a plot to vary independently for each row of facets

Graph to see if china and india are the drivers of asias growth

```
top_2_per_continent <- gdp_per_capita_df %>%
  group_by(Continent, Entity) %>%
  summarise(Total_GDP = sum(Gdp, na.rm = TRUE)) %>%
  arrange(Continent, desc(Total_GDP)) %>%
  group_by(Continent) %>%
  slice(1:2) %>%
  ungroup()

get_plot_data <- function(x) {

  top_2_entities <- top_2_per_continent %>%
    filter(Continent == x) %>%
    pull(Entity)

  country_data <- gdp_per_capita_df %>%
    filter(Continent == x, Entity %in% top_2_entities)

  continent_avg <- continents_growth_df %>%
    filter(Continent == x) %>%
    rename(Growth = Avg_Growth) %>%
    mutate(Entity = "Continent Average")
}
```

```

plot_data <- bind_rows(country_data, continent_avg)

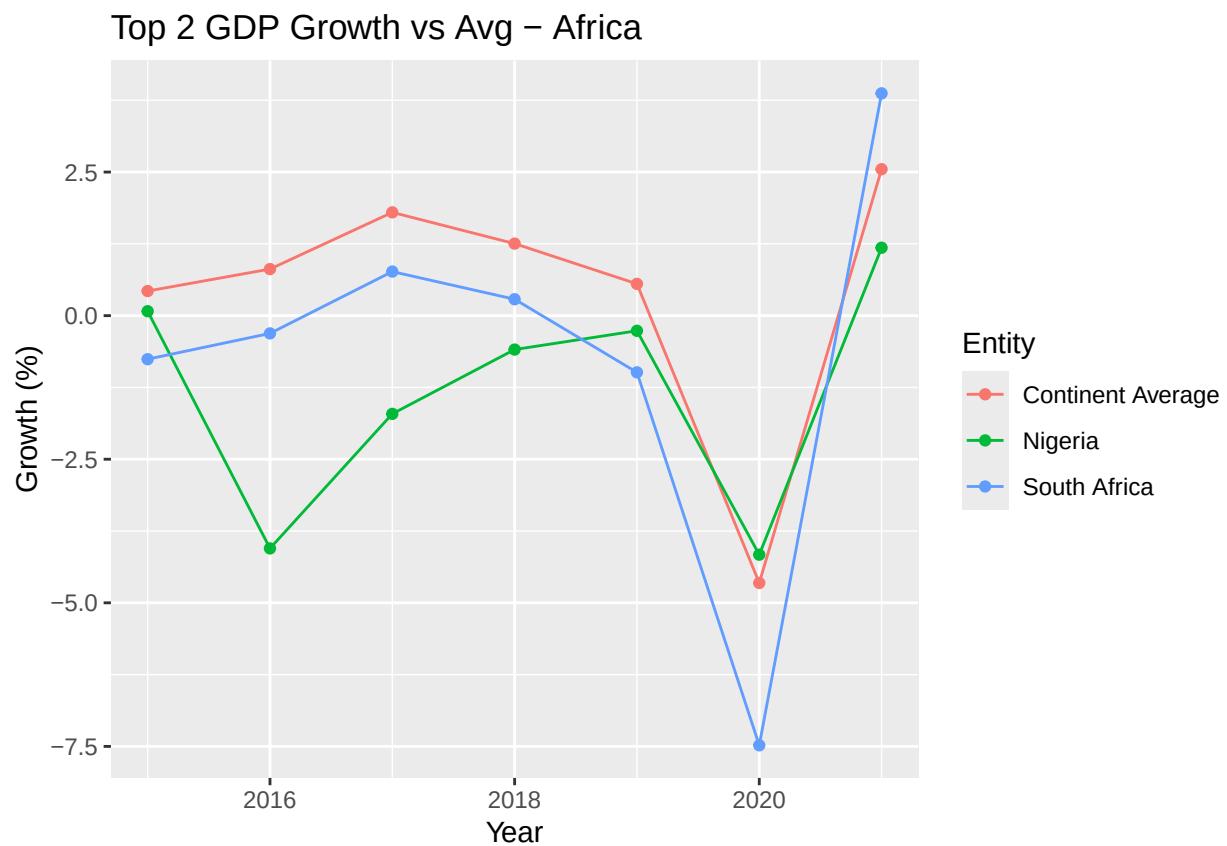
return(plot_data)
}

plot_drivers <- function(continent) {
  data <- get_plot_data(continent)

  ggplot(data, aes(x = Year, y = Growth, colour = Entity)) +
    geom_line() +
    geom_point() +
    labs(
      title = paste("Top 2 GDP Growth vs Avg -", continent),
      y = "Growth (%)"
    )
}

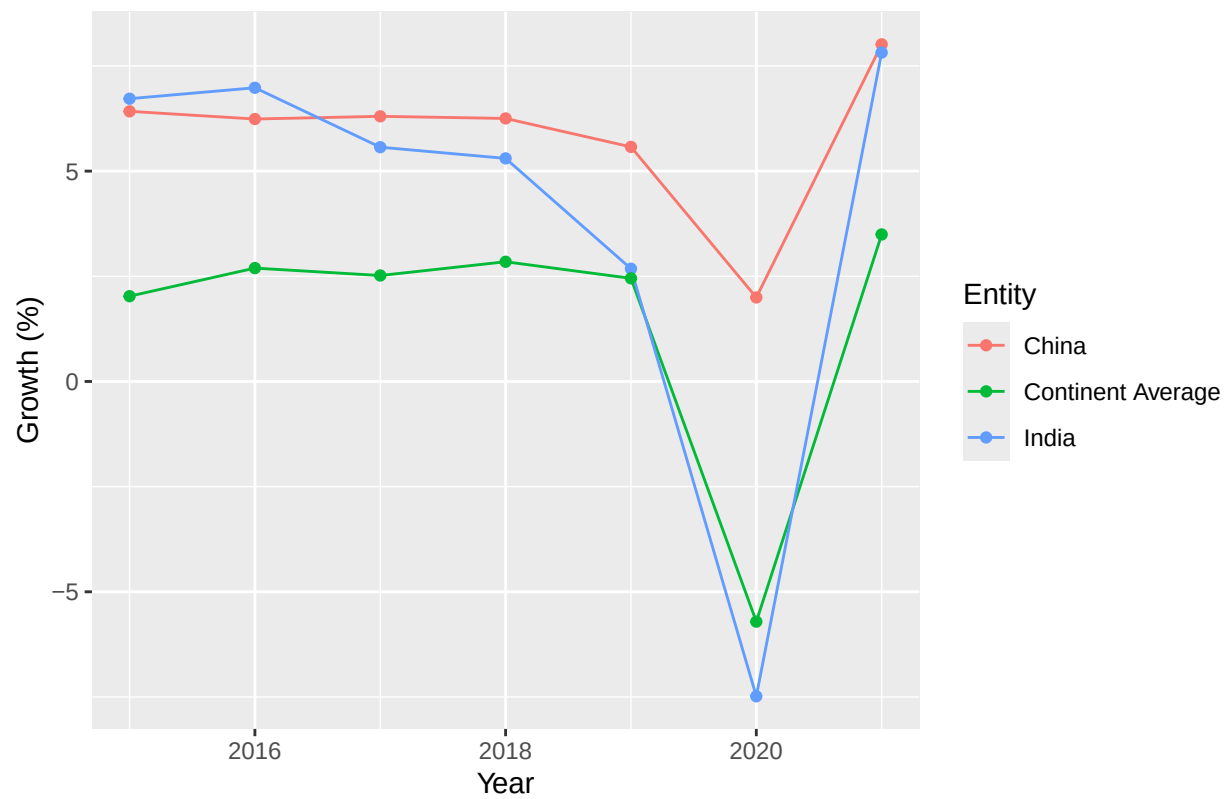
plot_drivers("Africa")

```



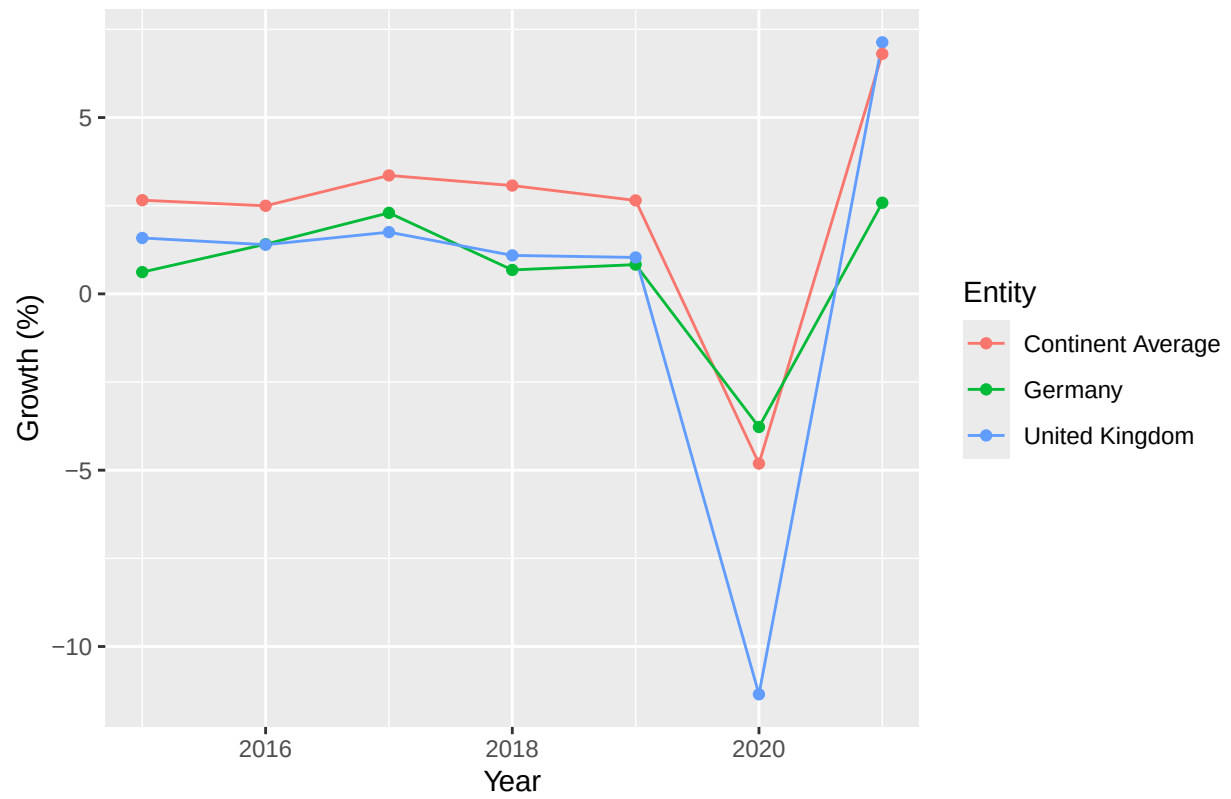
```
plot_drivers("Asia")
```

Top 2 GDP Growth vs Avg – Asia



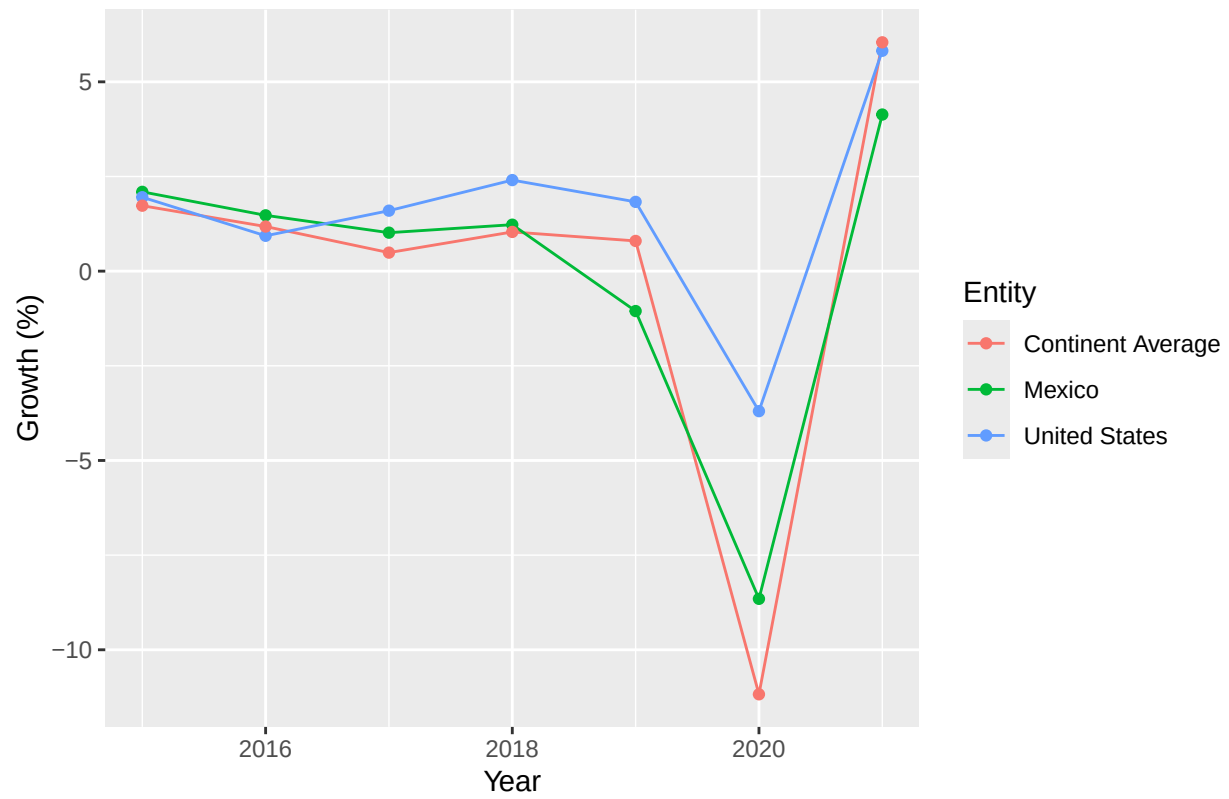
```
plot_drivers("Europe")
```

Top 2 GDP Growth vs Avg – Europe

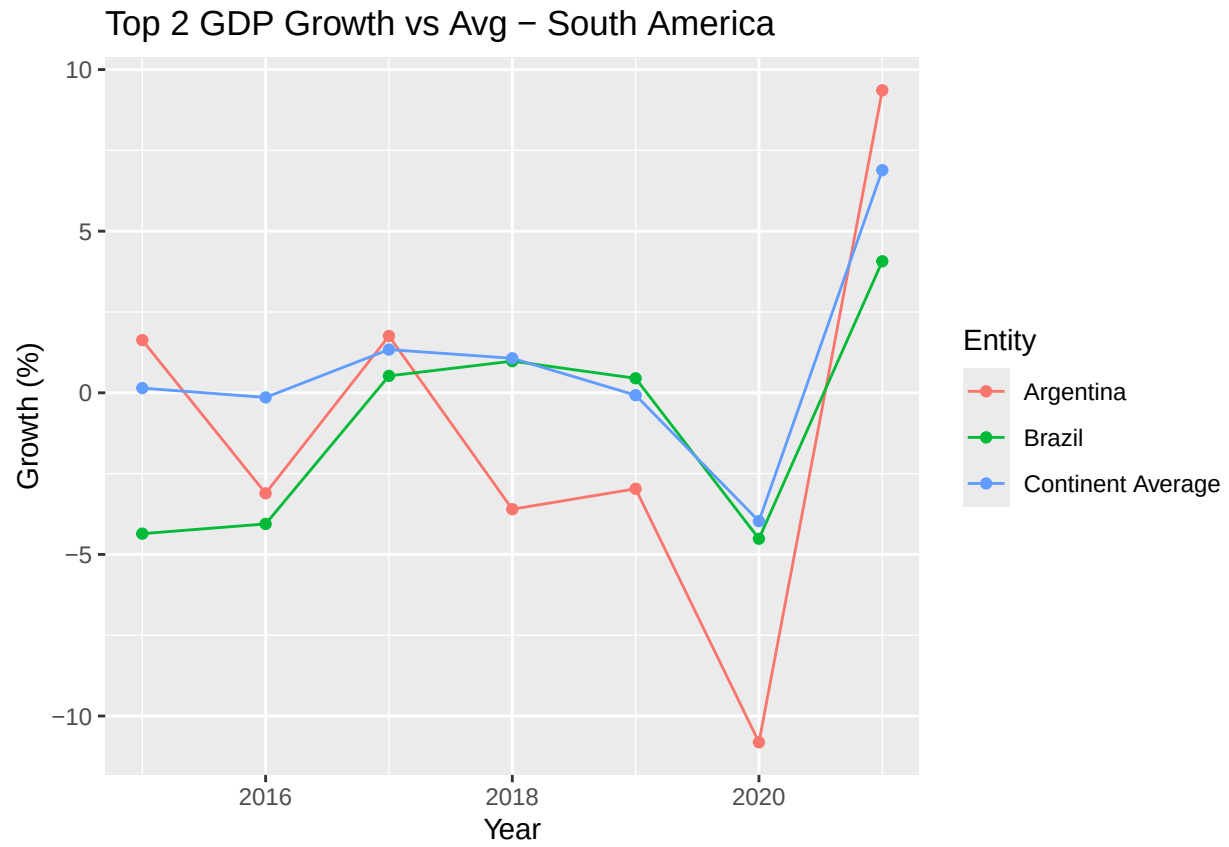


```
plot_drivers("North America")
```

Top 2 GDP Growth vs Avg – North America

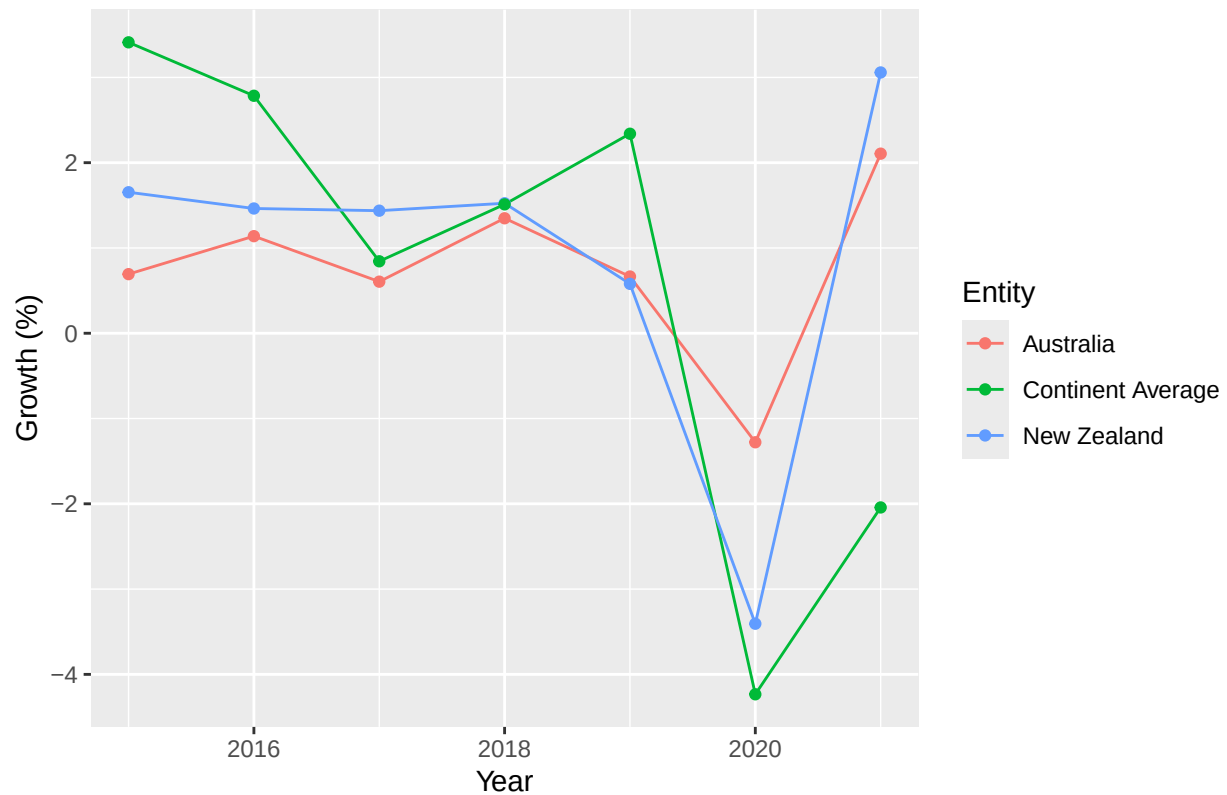


```
plot_drivers("South America")
```



```
plot_drivers("Oceania")
```

Top 2 GDP Growth vs Avg – Oceania



ANALYSIS Africa: Nigeria, South Africa, Algeria Asia: China, India, Japan Europe: Germany, UK, France. Germany leading North America: US, Mexico, Canada Oceania: Australia, New Zealand, Papa New Guinea South America: Brazil, Argentina, Colombia. Brazil drives volatility, steady increase

in all continents apart from Africa, there is one clear country that is driving GDP why this is important to analyse? these countries are the ones that drive continent verages the weighted averages are heavily influenced by large economics (eg US, China, Germany) spikes/dips in major economies explain most fluctations in continent graphs. this can definitely be seen in europe in thr last few years for example